

MedTriage-NLP

Symptom-to-Department Classification — Model Card

Overview

This model classifies a patient/case symptom description into one of 11 hospital departments (e.g., Cardiology, Orthopedics, ENT), to support faster and more consistent clinical triage/routing.

Dataset

Medical Transcriptions dataset (Kaggle, tboyle10) — sourced from mtsamples.com, released under CC0: Public Domain. After filtering to 11 clinically distinct, well-represented departments, 2,232 real case records were used.

Model Details

- Features: patient description + cleaned clinical keywords (department-name terms removed to avoid label leakage), vectorized using TF-IDF (unigrams + bigrams, sublinear term frequency).
- Classifier: Logistic Regression (C = 1, class-balanced weighting to handle department imbalance).
- Train / Test split: 80% / 20%, stratified by department (1,855 train / 377 test cases).

Overall Performance

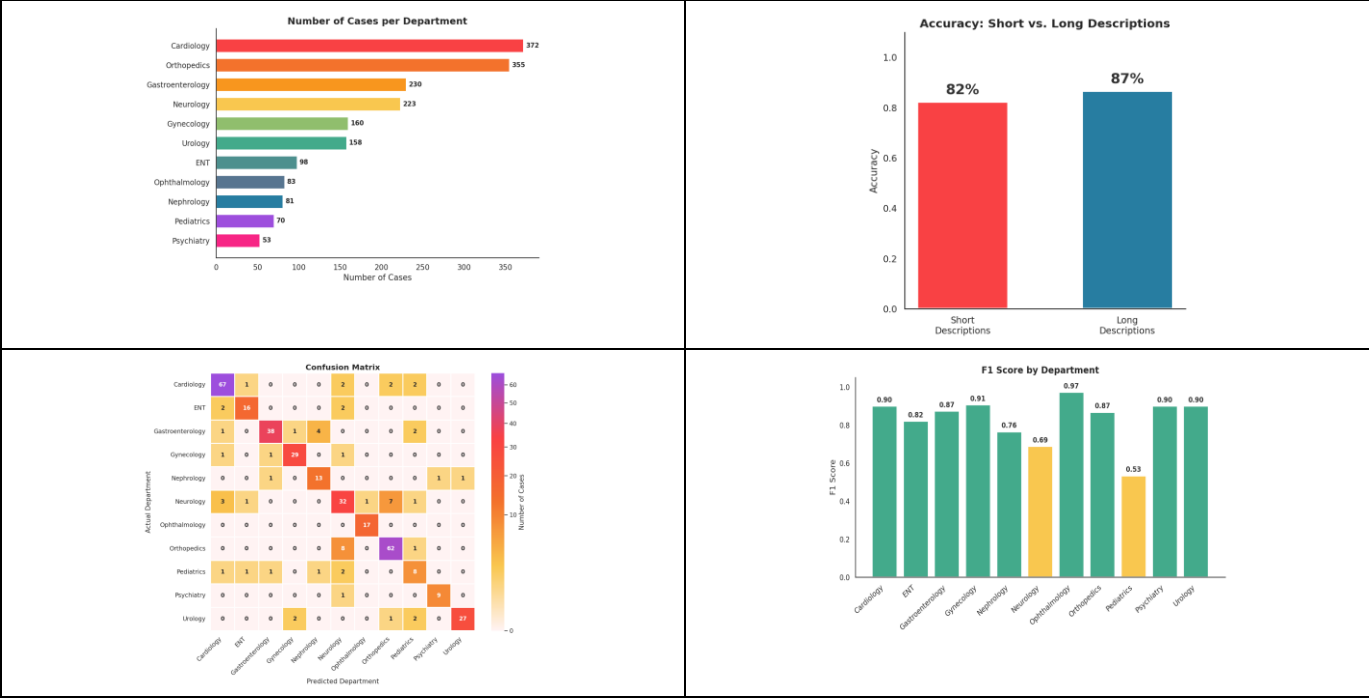
Metric	Score
Accuracy	84.3%
Macro F1-score	0.83
Weighted F1-score	0.85

Bias / Subgroup Audit

Test cases split at median word count (33 words) to compare reliability on brief vs. detailed descriptions:

Subgroup	Accuracy
Short descriptions (≤ 33 words)	82.3%
Long descriptions (> 33 words)	86.6%

Visual Summary



Explainability — Top Predictive Words

Department	Top Words
Cardiology	chest, cardiac, coronary, artery
Orthopedics	knee, fracture, joint, tendon
Ophthalmology	eye, vision, cataract, lid

Limitations

- Descriptions are clinician-authored summaries, not patient self-reported text — casual, real-world phrasing may reduce accuracy.
- Pediatrics (F1 = 0.53) and Neurology (F1 = 0.69) underperform due to smaller sample sizes and symptom vocabulary overlap.
- As a closed-set classifier, the model can only choose among its 11 trained departments.